



FAKULTI SAINS KOMPUTER DAN TEKNOLOGI MAKLUMAT
Faculty of Computer Science and Information Technology

WQF7007 NATURAL LANGUAGE PROCESSING
SEMESTER II SESSION 2025/2026

GROUP PROJECT:
TEXT-BASED MENTAL HEALTH DISORDER CLASSIFICATION USING NATURAL
LANGUAGE PROCESSING

OCCURRENCE:
OCC2

GROUP MEMBERS:

NAME	MATRIC NO
ANG ANSON	17203192
LIEW JIA YI	25075973
LIM CHIN PEI	24077115
YAP HUI QING	24080940
YAP JIUN SOONG	24070916
YEOH LI TING	23079401

LECTURER'S NAME:
DR. MOHAMED N. M. LUBANI

Introduction

The United Nations established Sustainable Development Goal 3 (SDG 3), good health and well-being, which aims to ensure healthy lifestyle and promote well-being for people. According to the World Health Organization (WHO), mental health is defined as a state of mental well-being that allows people to handle stress, learn effectively and perform well at work (Sohal & Sharma, 2025). Mental health consists of three components, including mental disorders, mental health problems and mental well-being (Rugulies et al., 2023). Clinical conditions like anxiety and depression are categorized as mental disorders. Mental health problems also include burnout and stress, which are not recognized as medical conditions. Nevertheless, mental well-being means a person has positive emotions, good mental functioning and healthy social relationships, rather than focusing on problems. Thus, maintaining mental well-being is essential for people to cope with life challenges and continue sustaining overall life satisfaction.

Globally, there are more than a billion people living with mental health conditions. At an early stage, mental health disorders are difficult to detect due to lack of awareness, stigma, limited to professional help and so on. People may fail to recognize early symptoms, resulting in worsening depression and anxiety. Mental health disorders also associated with symptoms, such as fatigue, sleep disturbances, weight loss and other related conditions (Zhou & Mohd, 2025). In addition, mental health disorders are affected by multiple factors, including economic, environmental and socioeconomic. For example, financial instability and unemployment can lead to increased stress. In terms of environmental aspect, individuals who are exposed to abuse, violence or toxic environments are more likely to face mental health disorders. For socioeconomic factors, poor interpersonal relationships also lead to stress and mental health issues. People could not receive early diagnosis and treatment due to limited access to education and healthcare services. As a result, effective approaches are needed to reduce the impact of mental health disorders.

In recent years, mental health detection has emerged as a key application of Natural Language Processing (NLP), focusing on detecting mental health conditions to support early diagnosis. The rise of social media platforms for people to express their emotions, thoughts and experience through text. Consequently, the text dataset provides valuable psychological insights. The advancement of deep learning has facilitated the development of transformer-based architecture in order to improve text classification. Bidirectional Encoder Representations from Transformers (BERT) is one of the transformer-based models that can achieve a deep understanding of the context by using the mechanism of bidirectional attention (Wagay et al., 2025). It provides the foundation for BERT variants, including DistilBERT, RoBERTa and ALBERT. Overall, BERT is suitable for mental health detection as it is able to capture better semantic relationships effectively.

Problem Statement

Large volumes of unstructured textual data are generated from social media, online forums, and other digital communications. Historically, traditional machine learning algorithms are implemented to detect mental health conditions, such as Naïve Bayes, Support Vector Machines (SVM). However, traditional machine learning heavily rely on manual feature engineering, which requires domain expertise to capture relevant information and patterns from textual data (Wagay et al., 2025). As a result, manual identification of mental health-related content is time-consuming,

inconsistent and not scalable. Furthermore, different linguistic patterns and user intent across social media platforms make it difficult to develop accurate model. Models often ignore important elements, such as emoji usage, slang, and emotional expressions in social media text (Zhou & Mohd, 2025). Therefore, the accuracy of mental health detection is affected by the challenges, such as limitations of traditional machine learning and the complexity of languages as well as unstructured textual data.

Objectives

- 1. To develop a text classification framework for identifying mental health categories from textual data.
- 2. To fine-tune a pre-trained BERT model for the mental health classification task.
- 3. To evaluate model performance using metrics such as accuracy, precision, recall and F1-score.

Methodology

Data Collection

The dataset used in this project is a mental health conditions classification data, available in CSV format and publicly accessible on the Hugging Face platform. It consists of 103,488 records with two primary features: text and status. The text column contains user-generated text content, while the status column represents labels corresponding to seven mental health conditions, namely normal, anxiety, depression, bipolar, stress, personality disorder and suicidal.

Tools and Environment

The project was implemented in Google Colab using Python. Hugging Face Datasets was used to load the dataset, pandas and visualization libraries were used for exploratory data analysis, Hugging Face Transformers was used for BERT tokenization and model development, and PyTorch served as the deep learning backend. The model was using the Hugging Face Trainer with GPU acceleration in Google Colab. Model performance was evaluated using accuracy, weighted precision, weighted recall, and weighted F1-score.

Exploratory Data Analysis

Table 1

Summary of Dataset Characteristics and Target Variable Distribution

Item	Result
Number of rows	103,488
Number of columns	2
Input column	text
Target column	status
Missing values	0
Duplicate rows	0
Number of classes	7

Target Variable	Number of Samples
anxiety	17,620
normal	16,068
depression	15,901
stress	15,230
personality disorder	13,915
bipolar	13,708
suicidal	11,046

The dataset contains no missing values and no duplicate rows. The target variable contains seven mental health categories, which are anxiety, bipolar, stress, personality disorder, normal, depression, and suicidal. The class distribution shows that the dataset is slightly imbalanced. The category with the highest number of samples is anxiety, with 17,620 samples, while the category with the lowest number of samples is suicidal, with 11,046 samples. Other categories include normal with 16,068 samples, depression with 15,901 samples, stress with 15,230 samples, personality disorder with 13,915 samples, and bipolar with 13,708 samples. Although the dataset is not perfectly balanced, the imbalance is not severe because the largest class is only about 1.6 times larger than the smallest class. Therefore, stratified data splitting and weighted evaluation metrics were used for this project.

Table 2
Descriptive Statistics and Text Length Analysis

Descriptive Statistic	Word Length (Words)
Number of rows	103,488
Mean	82.02
Median	34
25th percentile	24
75th percentile	99
Maximum	5,419

Text Length	Count	Percentage (%)
More than 128 words	16,367	15.82%
More than 256 words	6,600	6.38%
More than 512 words	1,614	1.56%

Before apply BERT tokenization, text length analysis was also performed to understand the length of the input text before applying BERT tokenization. Two new columns were created: `char_length`, which represents the number of characters in each text, and `word_length`, which represents the number of words in each text. The analysis showed that the average text length is approximately 82 words, while the median text length is 34 words. This means most of the text samples are relatively short. However, the maximum text length is 5,419 words, showing that some samples are extremely long. The text length distribution is right-skewed, meaning most texts are short, but a small number of texts are very long.

Table 2 results showed that 15.82% of the samples contain more than 128 words, 6.38% contain more than 256 words, and only 1.56% contain more than 512 words. Based on this result, a maximum sequence length of 256 tokens was selected for BERT tokenization. This value is suitable because it preserves most of the text information while keeping the training process manageable in Google Colab.

Data Preprocessing

Since the model used in this project is BERT, only light preprocessing was applied. This is because BERT uses contextual embeddings and subword tokenization, so heavy preprocessing techniques such as stopwords removal, stemming, and lemmatization were not used.

The preprocessing stage consisted of text cleaning, empty record removal, label encoding, dataset splitting, and conversion into Hugging Face dataset format. The original text data was cleaned using a simple text cleaning function. The function first converted each text value into string format to prevent data type errors. URLs were then removed because they do not provide useful semantic information for the classification task. Extra whitespace was also normalized, and leading or trailing spaces were removed. After applying the cleaning process, the dataset was checked for empty text records. One empty text record was identified after cleaning. This record was removed because empty text does not provide meaningful input for BERT and may negatively affect the training process. After removing the empty record, the final dataset used for modelling contained 103,487 samples.

Then workflow will proceed to encode categorical values in the dataset. Since the target variable consists of seven classes, label encoding will be used to convert the categorical label into numerical values.

Table 3

Encoding of Mental Health Categories to Numerical Label Identifiers

Label ID	Mental Health Category
0	anxiety
1	bipolar
2	depression
3	normal
4	personality disorder
5	stress
6	suicidal

Next, dataset was manually divided into training, validation, and test sets using an 80:10:10 ratio. The split was performed as follows:

Table 4

Distribution of Dataset Splits for Model Training and Evaluation

Dataset Split	Number of Samples (N)	Percentage (%)
Training set	82,789	80%
Validation set	10,349	10%

Test set	10,349	10%
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The training set was used to train the BERT model. The validation set was used to monitor model performance during training, while the test set was reserved for final model evaluation. Stratified splitting was applied using the encoded label column. This ensured that the proportion of each mental health category remained similar across the training, validation, and test sets. After splitting the dataset, the pandas DataFrames were converted into Hugging Face Dataset objects for efficient tokenization and training using the Hugging Face Trainer. BERT tokenization converts text into numbers that the BERT model can understand. It adds [CLS] at the beginning for classification and [SEP] at the end to mark the end of the text. The tokenizer creates input_ids, attention_mask, and token_type_ids.

Table 5

Structure and Functional Description of BERT Tokenizer Outputs

Tokenizer Output	Functional Description
input_ids	Numerical token IDs used by BERT
attention_mask	Indicates real tokens and padding tokens
token_type_ids	Segment IDs, mainly used for sentence-pair tasks

Texts longer than 256 tokens were truncated. Dynamic padding was used so that shorter texts were padded only during training, based on the longest text in each batch. This makes training more efficient because not every text needs to be padded to 256 tokens.

Model Architecture

The model training phase was conducted using the Hugging Face Trainer API in Google Colab. The pre-trained bert-base-uncased model was used as the base model and adapted for a seven-class mental health text classification task.

The BERT model was loaded using AutoModelForSequenceClassification, with a new classification layer added for the seven mental health categories. The training configuration used in this project included a learning rate of 2e-5 (0.00002), a batch size of 8, a weight decay of 0.01, and 3 training epochs. The model was evaluated after each epoch using the validation set. The best model was selected based on the weighted F1-score.

During training, the model processed the tokenized training data, generated predictions, calculated the training loss by comparing predictions with the true labels, and updated its weights through backpropagation. This allowed the pre-trained BERT model to adapt its language understanding to the mental health classification task.

Technical Stack

Table 6

Software Libraries and Frameworks Used for Data Processing and Model Development

Libraries / Tools	Functional Purposes
Pandas, Numpy	Dataset loading, manipulating and numerical operations.
Datasets (Hugging Face)	Fetching the dataset directly from Hugging Face Hub.

Matplotlib, Seaborn	Visualizing class distribution, text length and confusion matrix.
Regular Expression (re)	Text cleaning: URL and space removal.
Scikit-learn	Train-test splitting and evaluation metrics such as accuracy, precision, recall, and F1-score.
Transformers (Hugging Face)	Tokenizer, transformer-based sequence classification model, and training pipeline.
Pytorch	Using for deep learning framework.
Google Colab	Cloud-based environment for analysis and model training.
Streamlit / Flask	User interface for entering text and displaying mental health result.

Evaluation Strategy

Both models will be evaluated on the same held-out test set, which was not seen during training or validation. Four primary metrics are reported, following standard practice in NLP classification studies (Bokolo & Liu, 2023; Alhamed et al., 2024):

- Accuracy: The proportion of correctly classified samples across all classes.
- Precision: The proportion of positive predictions that are truly positive. Macro-averaging treats each class equally while weighted averaging accounts for class frequency.
- Recall: The proportion of actual positives correctly identified. Particularly important for minority classes.
- F1-score: The harmonic mean of precision and recall. Provides a single balanced measure, especially useful under class imbalance.

In addition to aggregate metrics, per-class precision, recall, and F1 are reported via a full classification report. Normalised confusion matrices are presented to visualise inter-class confusions.

Issues and Challenges

GPU Memory Constraints

Google Colab's free tier T4 GPU only provides 16 GB of VRAM. Hence, training bert base uncased requires careful memory budgeting. A maximum sequence length of 256 tokens and a batch size of 256 were chosen to remain well within the memory limit, maintaining a sufficient effective batch size.

Text Truncation

A certain subset of the corpus contains texts longer than 128 tokens. By truncating these sequences, it may discard diagnostic information present later in the text. To overcome this, EDA histogram was used to confirm that the majority of the informative content are captured within the first 128 tokens for this dataset.

Colab Session Timeouts

For a free Colab session, it will disconnect after approximately 90 minutes of inactivity and have a maximum runtime of about 4 hours. To ensure outputs are not lost, all the model checkpoints and visualizations are saved to google drive at end of each training phase to ensure no outputs are lost on disconnection.

Domain Shift and Generalisation

It is challenging to determine the source of the dataset. Thus, it would be safe to assume that it consists of synthetic or curated first person narratives. For the real-world deployment, this requires evaluation on the genuinely organic text, it could be from social media text, clinical notes or other sources where vocabulary, syntax, and topic distribution may differ significantly. (Zhang et al., 2022; Lee et al., 2020). This ensures more comprehensive representation during deployment.

Ethical Considerations

Automated mental health classifications raise significant ethical concerns. This includes privacy, stigmatization, and the risk of misdiagnosis. Deployment of such system must involve qualified mental health professionals in the decision phase to ensure appropriate interpretation and reduce the likelihood of incorrect assessment. In addition, consent must be obtained from users before collecting their data, ensuring that transparency and ethical responsibility throughout the process.

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